

CPH 200C PS3 · REGIONAL AI · 2026

Three Governing Systems for AI

How the US, Europe, and Asia deploy AI: policy, funding, technical developments, and landscape by region

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CPH 200C · Computational Precision Health

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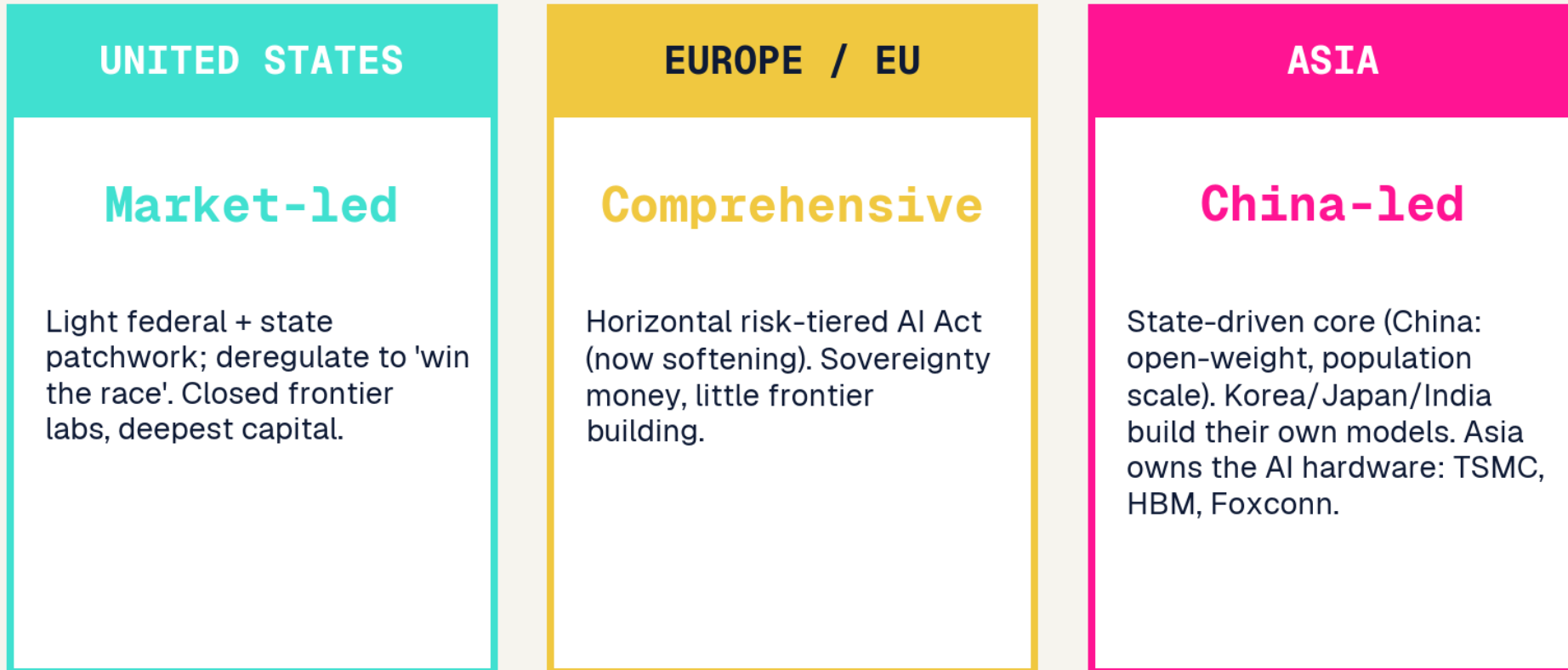


FRAMING

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Three governing systems for AI

Three regions, three governing systems for AI



Same technology, three different games. The rest of the deck is this slide, scope by scope.

Regulate, build, deploy

	UNITED STATES	EUROPE / EU	ASIA
REGULATE	Light, late, contested	Heaviest, first-mover	State-led to light (varies)
BUILD	Frontier leader	ASML chokepoint only	Fast follower + owns hardware
DEPLOY	Enterprise SaaS	Compliance-gated, slow	Population scale

What leaders predicted, and what's different now

CHINA

Kai-Fu Lee

AI Superpowers (2018)

The age of implementation: execution and data, not breakthroughs, decide the race, favoring China.

UNITED STATES

Eric Schmidt et al.

The Age of AI (2021); NSCAI

AI is a national-security race the US must win; warned of China closing the gap fast.

EUROPE / EU

Mario Draghi

EU Competitiveness Report (2024)

Europe regulates but does not build; structurally behind on compute, capital, and scale.

WHAT'S DIFFERENT NOW (2025):

None predicted a Chinese OPEN model (DeepSeek) would reset the frontier while US labs re-closed. Lee's implementation thesis holds, but the ROI gap shows capability is not yet value.



POLICY & REGULATION

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United States: executive whiplash plus a state patchwork

Executive whiplash

Trump rescinded Biden EO 14110 (Jan 2025); the July 2025 AI Action Plan deregulates to win the race; a Dec 2025 EO directs agencies to preempt state laws.

State patchwork

Colorado AI Act, Texas TRAIGA (Jan 2026), California SB 53, New York RAISE Act. A federal moratorium on state AI laws failed in the Senate 99 to 1.

Anti-woke procurement

EO 14319 bars federal purchase of LLMs that fail an ideological-neutrality test.

Frontier safety (Jun 2026)

Voluntary EO: labs grant up to 30-day pre-release access to NSA-designated frontier models + a Treasury cyber clearinghouse. No mandatory licensing, security-framed, not clinical.

Net effect

Light federal touch, a thickening state patchwork, and an open fight over who governs AI.

European Union: the AI Act, now softening

The AI Act

In force Aug 2024, phased: prohibited practices Feb 2025, foundation-model rules Aug 2025, high-risk 2026 to 2027. Penalties up to 35M euros or 7 percent of global turnover.

The softening

The Nov 2025 Digital Omnibus delays high-risk rules to 2027 and 2028 under competitiveness pressure; the separate AI Liability Directive was scrapped.

Regulates, does not build

Only 11 percent of EU firms use AI; 73 percent of foundation models are US-origin. Draghi: it is too late to build systemic challengers to US cloud.

Sovereignty money

InvestAI aims to mobilize 200B euros and build AI gigafactories, but ~90 percent of EU digital infrastructure is foreign-controlled.

China: vertical rules and alignment to state values

Vertical, not omnibus

No single AI law; iterative ministry rules. The 2023 Interim Measures govern public GenAI; 748 services were filed with regulators by Dec 2025.

Mandatory labeling

Since Sept 1 2025, all AI-generated content must carry explicit and hidden (metadata) labels.

Aligned to state values

Models must uphold core socialist values; China dropped a comprehensive AI law from its 2025 agenda in favor of pilots and standards.

Governance offensive

At the 2025 World AI Conference, Beijing proposed a World AI Cooperation Organization, courting the Global South against US go-it-alone policy.

Asia's regulatory split

China

State-vertical

Iterative ministry rules, AI-content labeling, alignment to state values. No omnibus law.

South Korea

Comprehensive

AI Basic Act (effective Jan 2026), the 2nd after the EU, but far lighter (fines cap ~\$21k).

Japan

Innovation-first

AI Promotion Act (2025): no penalties, guidance and name-and-shame only.

India

Light-touch

IndiaAI Governance Guidelines (Nov 2025); no standalone AI law, existing statutes adapted.

Singapore

Voluntary + tooling

Model AI Governance Framework + AI Verify, the first government AI-testing toolkit.

Regulatory philosophy, side by side

LIGHT-TOUCH

US, UK, Japan, India

Innovation first. Few or no penalties. Rules follow harm, if at all.

COMPREHENSIVE

EU, South Korea

One horizontal law, risk-tiered, ex-ante. Heaviest compliance load.

STATE-VERTICAL

China

Iterative sector rules + industrial policy + alignment to state values.

ASIA SPLITS

Japan: AI Promotion Act (2025), no penalties. South Korea: AI Basic Act (effective Jan 2026), EU-style comprehensive. India: light-touch + the IndiaAI Mission.

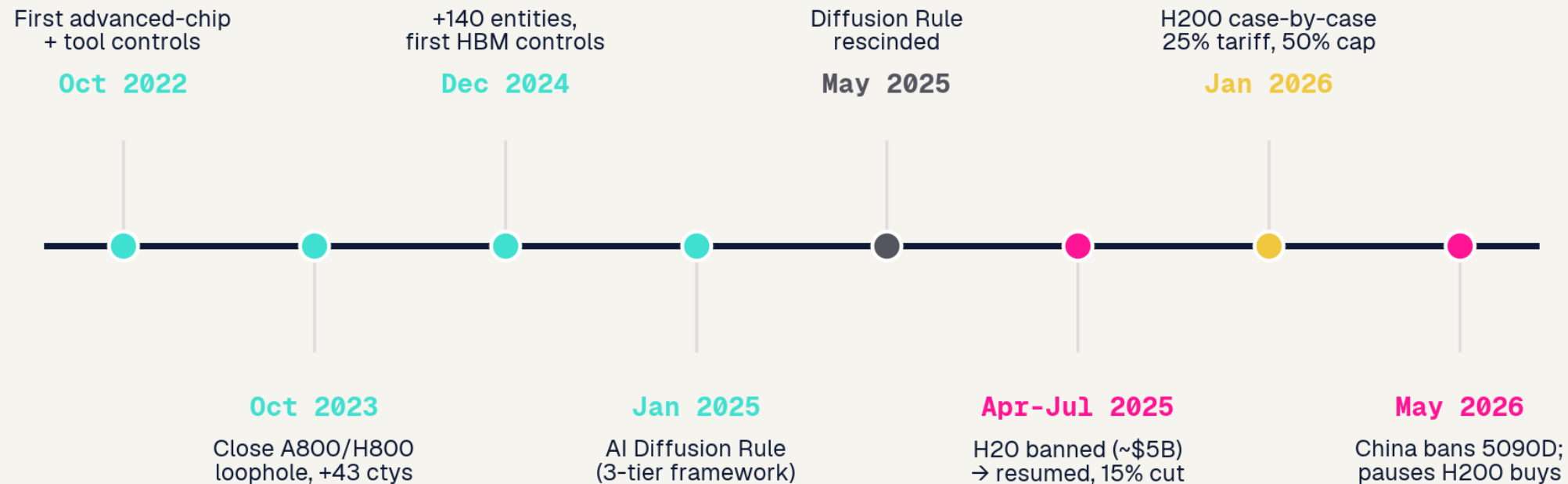


GEOPOLITICS & COMPUTE

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The chip war, 2022 to 2026

The chip war, 2022-2026: escalation, reversal, and China's pivot to domestic silicon



Sources: BIS / Commerce rules, NVIDIA filings, Tom's Hardware. The US tightened for three years, abruptly reversed in 2025, and China decoupled regardless.

The NVIDIA saga and China's domestic stack

EXPORT CONTROLS + NVIDIA

- H20 banned Apr 2025 (~\$5B NVIDIA charge), resumed Jul 2025 with a 15% cut of China revenue to the US government.
- H200 moved to case-by-case Jan 2026: 25% tariff, 50% volume cap, US testing.
- NVIDIA China share fell below 60%; the House AI Overwatch Act would bar Blackwell for two years.

CHINA'S DOMESTIC STACK

- Huawei Ascend 910C ~60% of an H100; CloudMatrix 384 links 384 chips to rival NVIDIA racks.
- Nine domestic chips put on the secure-and-reliable procurement list; state data centers must buy local.
- Chinese firms shipped 1.65M AI GPUs in 2025; China banned the 5090D and paused H200 buys.

Scale versus constraint: two paths to better models

UNITED STATES: SCALE

Answer the race with compute: Stargate (\$500B), ~\$700B hyperscaler capex in 2026, nuclear power deals. Frontier models stay closed. Abundant GPUs mean less pressure to economize.

Stargate; hyperscaler capex 2026

CHINA: CONSTRAINT

Capped on advanced GPUs by export controls, efficiency becomes the edge: mixture-of-experts, distillation, DeepSeek's ~\$6M training run, cheap inference. Necessity drives method innovation.

DeepSeek-V3 technical report, 2025

EUROPE: the upstream chokepoint. ASML's EUV monopoly is the gate the controls run through.

The provocative read: the GPU ceiling may be pushing China toward genuine methodology innovation, while the US mostly scales hardware. May 2026: Huawei's chairman publicly thanked the US for export controls, saying they 'supercharged' China's chip industry (Tom's Hardware; SCMP).

Compute and energy: the binding constraints

UNITED STATES

16 GW

data-center pipeline, but only ~5 GW under construction. ~\$700B hyperscaler capex in 2026. Power is the bottleneck.

CHINA

~400 GW

projected spare power by 2030 (added 430 GW wind+solar in 2025). 40 GW DC by end-2026. Energy is the advantage.

EUROPE / EU

EUR 200B

InvestAI + AI gigafactories. ASML holds the EUV monopoly. Compute and capital are the gap.

Asia owns the AI hardware

TSMC

Taiwan

Fabricates essentially all leading-edge AI chips. NVIDIA booked >60% of its CoWoS advanced-packaging capacity.

HBM

South Korea

SK Hynix holds ~62% of HBM memory and is NVIDIA's lead supplier; with Samsung, a global duopoly.

Foxconn

Taiwan

Builds >40% of the world's AI servers, the lead integrator for NVIDIA GB200/GB300 racks.

The hedge: sovereign AI means local-language models on foreign chips. Singapore runs Qwen; Malaysia runs DeepSeek on Huawei. Asia makes the hardware the world's AI runs on, but rents the models.



CORPORATE & MODELS

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United States: the closed frontier and a capital arms race

\$5.2T

NVIDIA market cap (world #1)

\$852B

OpenAI valuation

~\$965B

Anthropic (press-reported)

\$700B

2026 hyperscaler capex

The frontier shifted back to CLOSED: after Llama 4 stumbled and DeepSeek leveraged its open weights, even Meta pivoted toward proprietary models. OpenAI, Anthropic, and Google keep their best models closed to recoup training cost and limit misuse.

Europe: one chokepoint, a thin application layer

ASML

The one irreplaceable asset. Monopoly on EUV lithography (~90% share, ~\$250M/machine); EUR 32.7B 2025 sales. Every advanced NVIDIA or TSMC chip depends on it. Europe's leverage is hardware, not models.

Mistral (FR)

EUR 11.7B; ASML took ~11%

Helsing (DE)

defense AI, EUR 12B

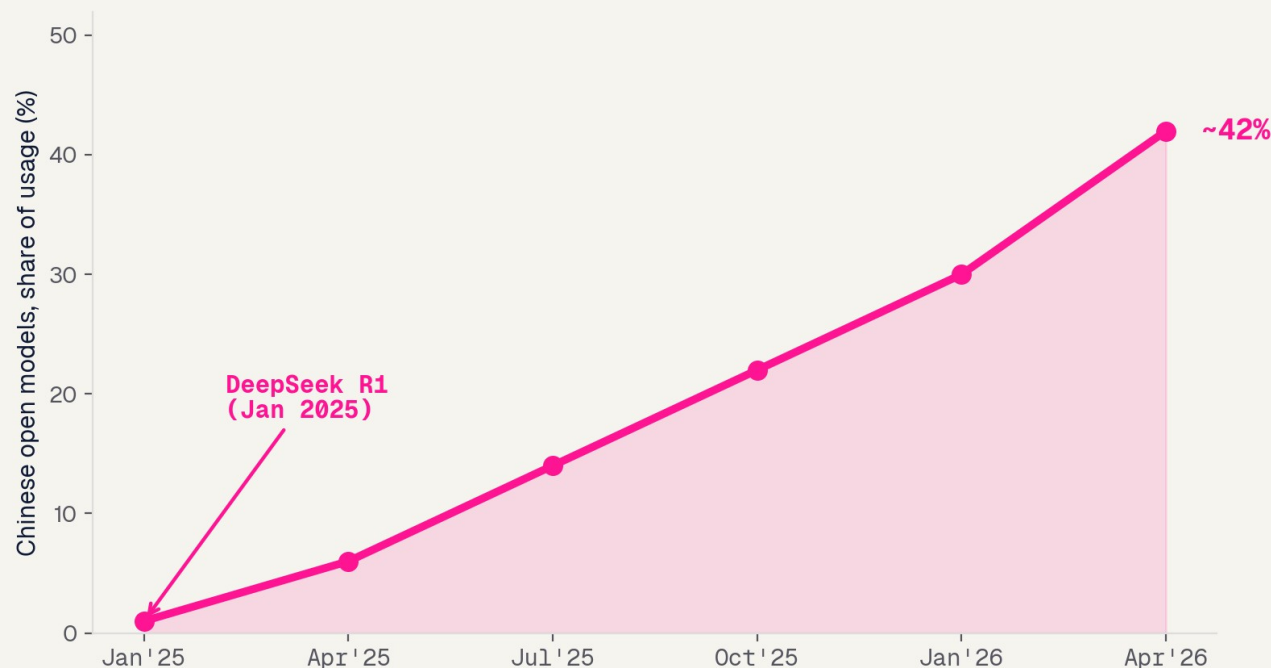
ElevenLabs / Synthesia

\$11B / ~\$4B; application layer

China: the open-weight surge

China's open-weight surge

Open models (DeepSeek, Qwen) rose from ~1% to ~40%+ of OpenRouter usage in a year; HF downloads China 17.1% > US 15.9%



The DeepSeek moment

R1 (Jan 2025), trained for a reported ~\$5.6M, topped the US App Store and wiped ~\$600B off NVIDIA in a day.

An open-weight ecosystem

Qwen, Ernie, Kimi, GLM, Doubao, Pangu. Price wars (Qwen cut up to 97%); Doubao serves >120T tokens/day.

Global South adoption

Singapore's government chose Qwen; Malaysia's stack runs on DeepSeek. Open weights are becoming an export.

Asia beyond China

INDIA

POPULATION-SCALE

Sarvam (govt-backed sovereign LLM) + Krutrim, under the national IndiaAI Mission. AI layered on UPI / Aadhaar at population scale; Qure.ai screens TB nationally.

IndiaAI Mission; PIB 2025

SOUTH KOREA

CLINICAL-AI EXPORTER

Naver HyperCLOVA X, LG EXAONE (open-weight), Samsung Gauss, backed by a national AI fund. Lunit + VUNO export FDA-cleared clinical AI worldwide.

MSIT; TechCrunch 2025

JAPAN

INNOVATION-FIRST

Sakana AI, the country's top unicorn; SoftBank owns ~11% of OpenAI. No-penalty AI law; AI-hospital pilots; Rapidus chasing a 2nm fab.

AI Promotion Act 2025

SOUTHEAST ASIA

THE BATTLEGROUND

SEA-LION switched from Meta's Llama to Alibaba's Qwen; Sahabat-AI (Indonesia), FPT (Vietnam). US and Chinese models compete for the next billion users.

TechNode 2025

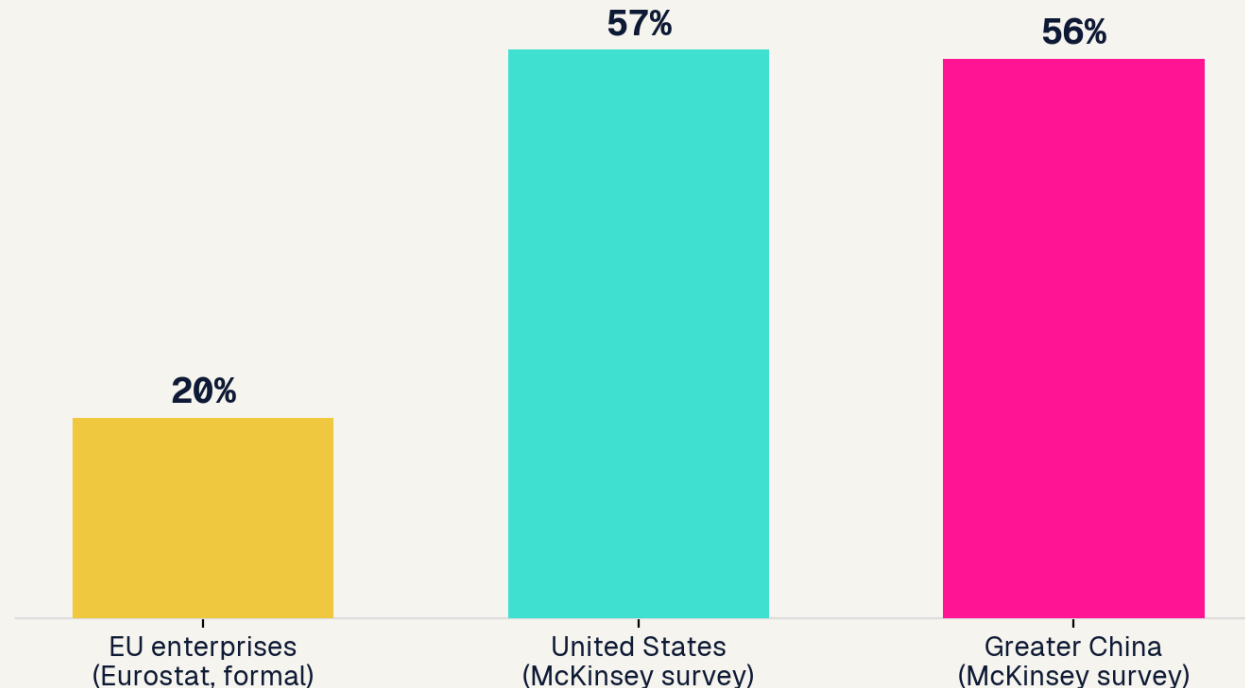


DEPLOYMENT & ADOPTION

Who actually uses AI

Who is actually using AI

US & China near-parity; EU lags. NB: Eurostat (formal adoption) and McKinsey (broad survey) use different definitions.



Eurostat measures formal adoption; McKinsey is a broad survey. Different definitions, same conclusion: the EU lags.

Near-parity on use

McKinsey: 88% of organizations use AI somewhere, but only ~1/3 have scaled it and 39% see real EBIT impact.

Consumer

ChatGPT ~900M weekly users, ~2.5x Gemini. China runs on its own: Doubao ~155M, DeepSeek ~82M. Western apps are absent there.

How AI gets deployed

CHINA

Super-app + state

AI agents inside WeChat; agentic commerce; state-coordinated, hardware-integrated. Deployment is centralized and consumer-facing.

UNITED STATES

Enterprise SaaS

AI flows through the vendor layer (Microsoft, Google, Salesforce). Specialized-vendor buys succeed ~67% vs ~1/3 for internal builds.

EUROPE / EU

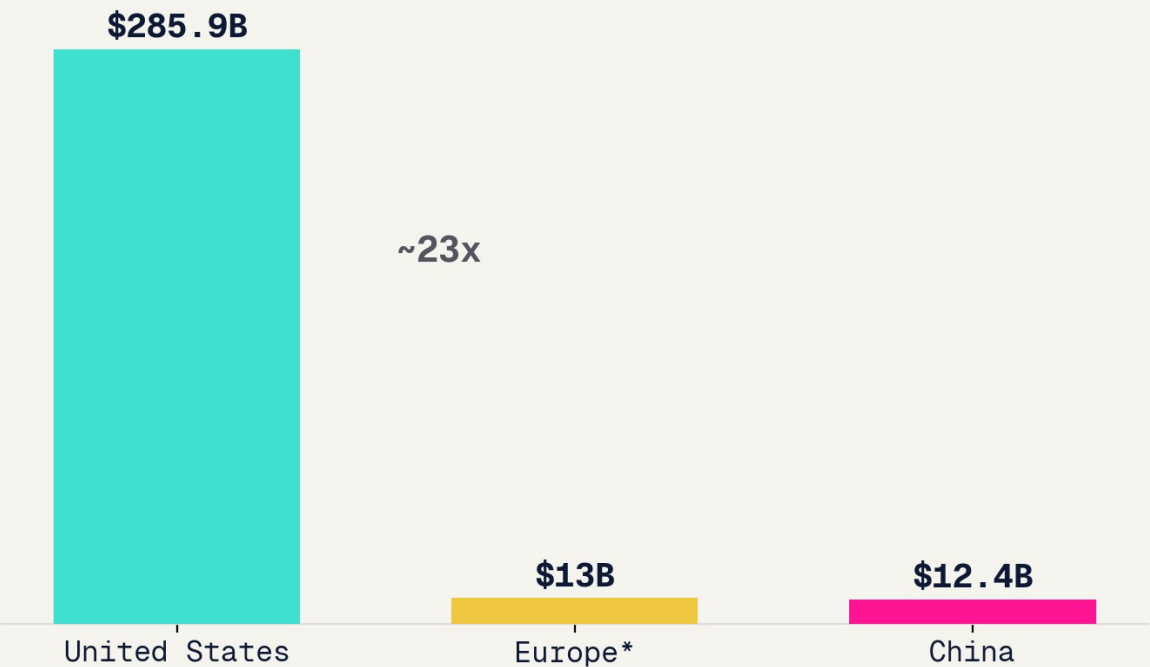
Compliance-gated

~60% of EU/UK developers report AI Act launch delays; deployment is slowed by the rulebook it wrote.

The capability scoreboard and the ROI gap

Private AI investment, 2025

Stanford AI Index (US, China). *Europe = est. EU+UK AI VC. The US captured ~83% of global private AI investment.



Models (2025)

US ~50 / China ~30 notable models

Frontier

Opus 4.8 / GPT-5.5 / Gemini 3.1 lead; DeepSeek V4 near-frontier at ~1/10 cost (US still on top)

Patents

China holds ~70% of granted AI patents

Investment

US ~83% of global private AI funding

The ROI gap: MIT found 95% of enterprise generative-AI pilots delivered no measurable P&L impact, despite \$30-40B spent. Capability and capital are not the same as realized value.



STARTUPS & SUPPORT

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Capital and leniency

\$222B

US AI startup funding (65% of all US VC)

state

China private VC fell <25%; guidance funds filled in (140+ gov deals)

gap

Europe: structural late-stage capital gap

LENIENCY, most permissive to ship fast:

US

China

UK

EU

permissive -----> constraining (EU sandboxes now delayed to Aug 2027)

Government support

UNITED STATES

\$500B Stargate (privately financed) + CHIPS Act spillover. Light direct funding, heavy private capital.

EUROPE / EU

InvestAI EUR 200B; France alone pledged EUR 109B at the Paris summit, the largest single-event government commitment.

UNITED KINGDOM

AI Growth Zones, a Sovereign AI Unit up to GBP 500M, and a pledge to grow sovereign compute 20x by 2030.

CHINA

National VC Guidance Fund (~\$143B over 20 yrs); the guidance-fund system runs ~2,178 funds worth ~\$900B, plus procurement preference.

Talent, immigration, and founder-friendliness

UNITED STATES: drawbridge UP

A \$100,000 per-petition H-1B fee took effect Sept 2025, a startup tax. ~60% of top US AI startups had an immigrant founder; ~70% arrived on student visas.

CHINA: drawbridge DOWN

The new K-visa (Oct 2025) is uncapped, 5-year, no employer sponsor, aimed at STEM/AI. Returnee sea-turtles anchor frontier labs, timed to exploit US restrictions.

Founder-friendliness: US = capital + market, rising talent cost. China = state capital + deployment speed, but directed + walled. EU = subsidy + legal certainty, heaviest compliance. UK = most balanced.

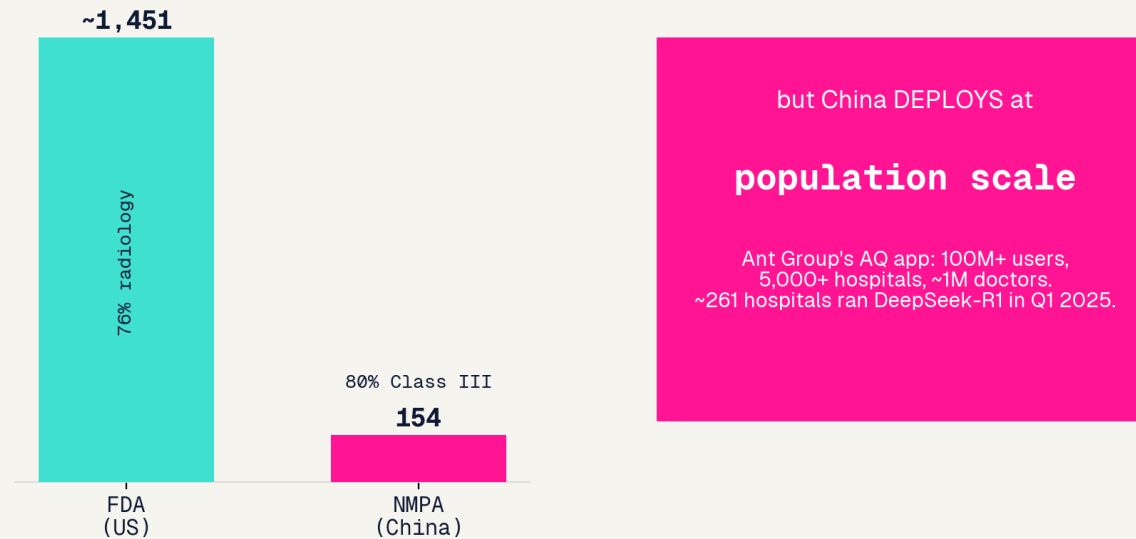


THE HEALTH THREAD

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Clinical AI by region

Clinical AI: the US approves the most discrete tools; China deploys at the most scale
Cumulative AI device approvals



United States

~1,451 AI devices cleared, 76% radiology, but reimbursement-starved and no pathway for generative/LLM clinical AI.

Deployed: Aidoc (1,600+ centers, 100M+ cases), Viz.ai stroke (1,800 hospitals).

Europe / EU

Medical AI is high-risk under the AI Act, stacked on MDR/IVDR, a double conformity burden; the health-data space is deferred to 2029.

Deployed: Oxipit ChestLink, world-first fully autonomous radiology AI (CE IIb).

China

Fewer approvals (154, 80% Class III) but deploys at population scale, state-coordinated.

Deployed: Infervision ~20k lung scans/day; Ant AQ 100M users; DeepSeek-R1 in 261 hospitals.

Same scale, different layer of the stack

UNITED STATES

Around the core

LLMs at scale, but OUTSIDE diagnosis: ambient scribes, medical coding, prior authorization, patient messaging. No FDA pathway for generative diagnosis, so AI stays off the diagnostic decision.

Abridge; Nuance DAX; Epic

CHINA

Inside the core

Clinical LLMs straight into diagnostic reasoning and the EHR: DeepSeek-R1 across 261 hospitals; Ant AQ reaches 100M users and ~5,000 hospitals.

Nature Medicine 2025

EUROPE / EU

No human, but only the easy cases

The only fully AUTONOMOUS diagnosis, used pragmatically: Oxipit ChestLink auto-signs off NORMAL chest X-rays (the ~15-40% with no finding) and routes every positive to a radiologist. CE Class IIb, a clearance the FDA has not granted.

Oxipit / Sectra, 2026

Same massive scale, three bets on WHERE AI sits: the US automates the paperwork around the clinician, China augments the diagnostic reasoning, Europe removes the human, but only on the cases that are clearly normal.

The regulate, build, deploy split, in medicine

UNITED STATES

APPROVES most

~1,451 discrete tools, device-by-device 510(k). But reimbursement is the bottleneck and LLM clinical AI has no pathway.

EUROPE / EU

REGULATES hardest

AI Act + MDR/IVDR double conformity, notified-body bottleneck. Germany's DiGA is the clearest working reimbursement route.

CHINA

DEPLOYS most

Population-scale platforms (AQ 100M users); 261 hospitals ran DeepSeek-R1. State coordination + volume procurement drive scale.



SYNTHESIS

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Beyond the big three

THE GULF: capital

Sovereign money, not models. Stargate UAE (5 GW; G42, MGX), Saudi HUMAIN, and MGX committed up to \$500B to US Stargate. Abundant capital and compute, thin domestic talent and market.

THE GLOBAL SOUTH: the battleground

Where US and Chinese open models compete for the next billion users. Sovereign-language models (SEA-LION, Sahabat, TAIDE) run on rented compute, increasingly on Chinese open weights.

Three hubs build and govern AI. Everyone else deploys it, increasingly on Chinese open weights, which is itself a geopolitical outcome.

The master comparison

Scope	United States	Europe / EU	Asia (China-led)
Governance	Market-led, light federal + states	Comprehensive AI Act (softening)	State-driven (China); mixed across Asia
Frontier models	Dominant, closed	Thin (Mistral; DeepMind under Google)	Open-weight (China); KR/JP/IN building
Capital	~83% of global private AI \$	Late-stage capital gap	State guidance funds; Gulf money
Compute	Power-constrained	One chokepoint (ASML)	Owns the hardware: TSMC, HBM, Foxconn
Deployment	Enterprise SaaS	Compliance-gated	Super-app, population scale
Clinical AI	Most approvals, underpaid	Most regulated	Most deployed (China); Korea exports
Founder-friendly	Capital + speed; talent cost up	Subsidy + certainty; heavy rules	State capital (China); open (India/SEA)

Takeaways and outlook

01

No region does all three well. The US builds the frontier, China deploys at scale, the EU regulates, and each is winning a different race.

02

The structural divergence is open vs closed. China's open-weight surge is reshaping who deploys AI across the Global South, not just at home.

03

Constraints differ by region: US power, EU capital + its own rules, China advanced chips, but China's domestic stack and energy surplus are closing the gap.

04

In medicine the same split holds: US approves most, EU regulates hardest, China deploys most. The deployed-AI bottleneck is institutional, not just technical.

References

Topic	Source
US AI Action Plan; preemption EO	White House (Jul 2025, Dec 2025)
EU AI Act + Digital Omnibus	White & Case; European Parliament (2025-26)
China GenAI + labeling rules	China-Briefing; Loeb (2023-25)
Chip controls; H20/H200; revenue-share	BIS; PBS; Tom's Hardware (2022-26)
China domestic chips; CloudMatrix	SemiAnalysis; Tom's Hardware (2025-26)
Adoption; ROI gap (95% pilots)	McKinsey; Eurostat; MIT/Fortune (2025)
Open-weight share; models; patents	MIT Tech Review; Stanford HAI (2025-26)
Investment; VC; valuations	Stanford AI Index; PitchBook (2025-26)
Talent: H-1B fee; China K-visa	Bulletin of Atomic Scientists; C&EN (2025)
Clinical AI: FDA / EU / NMPA	npj Digital Medicine; JMIR; Reed Smith (2025-26)

Full citations with URLs in the companion research.md.

Thank You

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夢想